ASSIGNMENT 14.1

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Batch-27

QUESTION-1

You are building a personal portfolio website to showcase your work.

Requirements:

• Create sections for About Me, Projects, and Contact.

• Use AI to:

o Suggest color palettes and typography.

o Create a responsive layout with Grid/Flexbox.

o Add smooth scrolling navigation

CODE:

index.html > html > head > style > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Personal Portfolio</title>
  <!-- AI-suggested color palette and typography -->
  <style>
    :root {
      --primary: #2D3142;
      --accent: #EF8354;
      --background: #F5F5F5;
      --text: #4F5D75;
      --heading: #1A1A2E;
      --font-main: 'Montserrat', Arial, sans-serif;
      --font-heading: 'Poppins', Arial, sans-serif;
    }
    html {
      scroll-behavior: smooth;
      background: var(--background);
      color: var(--text);
      font-family: var(--font-main);
    }
    body {
      margin: 0;
      padding: 0;
    }
    header {
      background: var(--primary);
      color: #fff;
      padding: 1rem 0;
      position: sticky;
      top: 0;
      z-index: 100;
    }
    nav {
      display: flex;
```

index.html

index.html > html > head > style > html

```html
<html lang="en">
<head>
  <style>
    nav {
      justify-content: center;
      gap: 2rem;
    }
    nav a {
      color: #fff;
      text-decoration: none;
      font-weight: 600;
      font-family: var(--font-heading);
      transition: color 0.3s;
    }
    nav a:hover {
      color: var(--accent);
    }
    main {
      display: grid;
      grid-template-columns: 1fr;
      gap: 2rem;
      max-width: 900px;
      margin: 2rem auto;
      padding: 0 1rem;
    }
    section {
      background: #fff;
      border-radius: 8px;
      box-shadow: 0 2px 8px rgba(45,49,66,0.08);
      padding: 2rem;
    }
    h2 {
      font-family: var(--font-heading);
      color: var(--heading);
      margin-top: 0;
    }
    /* Responsive layout */
```

index.html

index.html > html > head > style > html

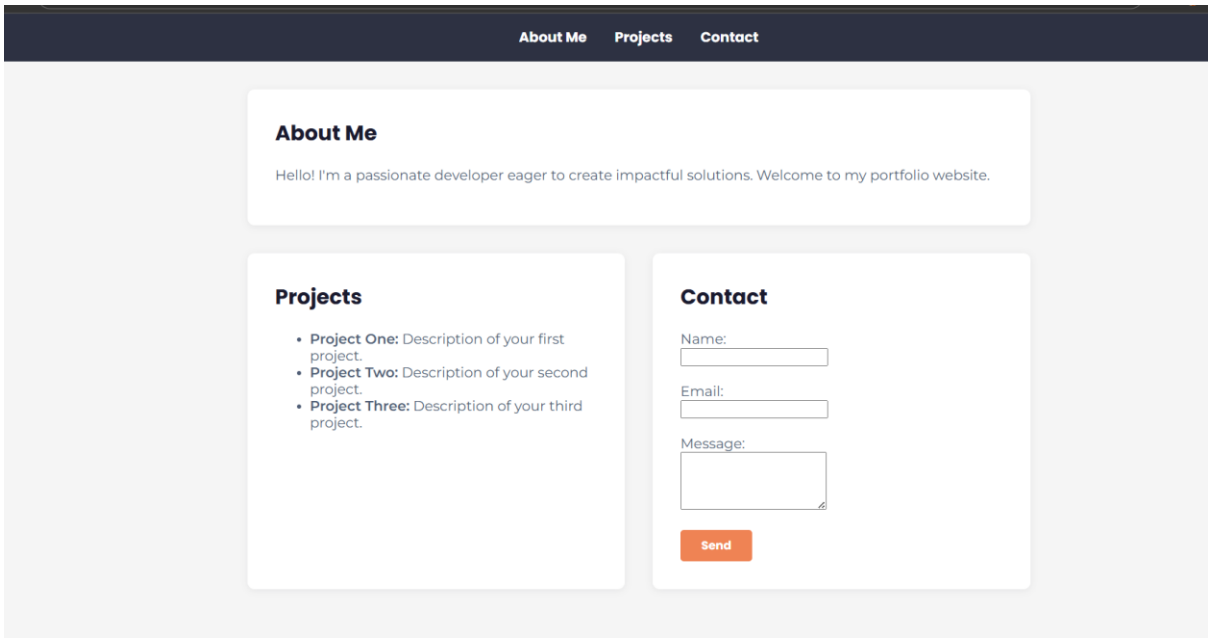
```html
<html lang="en">
<head>
  <style>
    /* Responsive layout */
    @media (min-width: 700px) {
      main {
        grid-template-columns: repeat(2, 1fr);
      }
      #about {
        grid-column: 1 / 3;
      }
    }
    @media (max-width: 699px) {
      main {
        display: flex;
        flex-direction: column;
      }
    }
  </style>
  <!-- Google Fonts -->
  <link href="https://fonts.googleapis.com/css?family=Montserrat:400,600&display=swap" rel="stylesheet">
  <link href="https://fonts.googleapis.com/css?family=Poppins:700&display=swap" rel="stylesheet">
</head>
<body>
  <header>
    <nav>
      <a href="#about">About Me</a>
      <a href="#projects">Projects</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>
  <main>
    <section id="about">
      <h2>About Me</h2>
      <p>Hello! I'm a passionate developer eager to create impactful solutions. Welcome to my portfolio website.</p>
    </section>
    <section id="projects">
```

index.html

index.html > html > head > style > html

```html
<html lang="en">
<body>
  <main>
    <section id="projects">
      <h2>Projects</h2>
      <ul>
        <li><strong>Project One:</strong> Description of your first project.</li>
        <li><strong>Project Two:</strong> Description of your second project.</li>
        <li><strong>Project Three:</strong> Description of your third project.</li>
      </ul>
    </section>
    <section id="contact">
      <h2>Contact</h2>
      <form>
        <label for="name">Name:</label><br>
        <input type="text" id="name" name="name" required><br><br>
        <label for="email">Email:</label><br>
        <input type="email" id="email" name="email" required><br><br>
        <label for="message">Message:</label><br>
        <textarea id="message" name="message" rows="4" required></textarea><br><br>
        <button type="submit" style="background: var(--accent); color: #fff; border: none; padding: 0.5rem 1.5rem; bor
      </form>
    </section>
  </main>
</body>
</html>
```

OUTPUT:



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QUESTION 2

Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

• Display product image, title, price, and "Add to Cart" button.

• Use AI to:

o Style with BEM methodology.

o Make layout responsive.

o Add hover effects and "Add to Cart" alert

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CODE

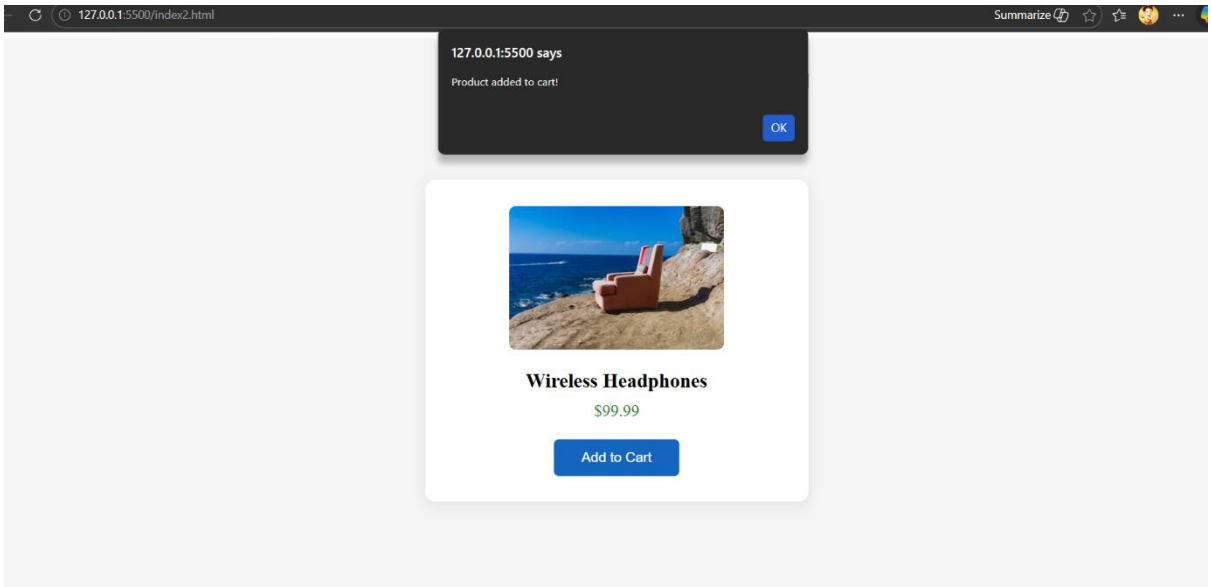
```html
index2.html > ...
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Online Store Product Page</title>
    <style>
        /* BEM methodology */
        .product-page {
            display: flex;
            justify-content: center;
            align-items: center;
            min-height: 100vh;
            background: #f5f5f5;
        }
        .product-page__container {
            background: #fff;
            border-radius: 12px;
            box-shadow: 0 4px 16px rgba(0,0,0,0.08);
            padding: 32px;
            max-width: 400px;
            width: 100%;
            display: flex;
            flex-direction: column;
            align-items: center;
        }
        .product-page__image {
            width: 100%;
            max-width: 260px;
            border-radius: 8px;
            margin-bottom: 24px;
            transition: box-shadow 0.3s;
        }
        .product-page__image:hover {
            box-shadow: 0 8px 24px rgba(0,0,0,0.15);
        }
```

```styl
        .product-page__title {
            font-size: 1.5rem;
            font-weight: 600;
            margin-bottom: 12px;
            text-align: center;
        }
        .product-page__price {
            font-size: 1.25rem;
            color: #2e7d32;
            margin-bottom: 24px;
        }
        .product-page__add-to-cart {
            background: #1976d2;
            color: #fff;
            border: none;
            border-radius: 6px;
            padding: 12px 32px;
            font-size: 1rem;
            font-weight: 500;
            cursor: pointer;
            transition: background 0.3s, transform 0.2s;
        }
        .product-page__add-to-cart:hover {
            background: #1565c0;
            transform: scale(1.05);
        }
        @media (max-width: 600px) {
            .product-page__container {
                padding: 16px;
                max-width: 100%;
            }
            .product-page__image {
                max-width: 100%;
            }
```

:

```html
<head>
    <style>
        @media (max-width: 600px) {
            .product-page__container {
                max-width: 100%;
            }
            .product-page__image {
                max-width: 100%;
            }
        }
    </style>
</head>
<body>
    <div class="product-page">
        <div class="product-page__container">
            <img class="product-page__image" src="https://images.unsplash.com/photo-1519125323398-675f0ddb6308?auto=form
            <div class="product-page__title">Wireless Headphones</div>
            <div class="product-page__price">$99.99</div>
            <button class="product-page__add-to-cart" onclick="addToCartAlert()">Add to Cart</button>
        </div>
    </div>
    <script>
        function addToCartAlert() {
            alert('Product added to cart!');
        }
    </script>
</body>
</html>
```

OUTPUT:



QUESTION 3

Code:

```
        name.addEventListener('input', validateName);
        email.addEventListener('input', validateEmail);
        phone.addEventListener('input', validatePhone);
        session.addEventListener('change', validateSession);

        form.addEventListener('submit', function(e) {
            let valid = true;
            if (!validateName()) valid = false;
            if (!validateEmail()) valid = false;
            if (!validatePhone()) valid = false;
            if (!validateSession()) valid = false;
            if (!valid) {
                e.preventDefault();
            }
        });
    </script>
</body>
</html>
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Conference Event Registration</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background: #f4f6f8;
            margin: 0;
            padding: 0;
        }
        .container {
            max-width: 420px;
            margin: 40px auto;
            background: #fff;
            border-radius: 8px;
            box-shadow: 0 2px 8px rgba(0,0,0,0.08);
            padding: 32px 24px;
        }
        h2 {
            text-align: center;
            margin-bottom: 24px;
            color: #2c3e50;
        }
        label {
            display: block;
            margin-bottom: 6px;
            font-weight: 500;
        }
        input, select {
            width: 100%;
            padding: 10px;
            margin-bottom: 18px;
            border: 1px solid #ccd0d5;
            border-radius: 4px;
```

```html
        input, select {
            font-size: 16px;
        }
        input:focus, select:focus {
            border-color: #2980b9;
            outline: none;
        }
        .error {
            color: #e74c3c;
            font-size: 14px;
            margin-top: -14px;
            margin-bottom: 10px;
        }
        button {
            width: 100%;
            padding: 12px;
            background: #2980b9;
            color: #fff;
            border: none;
            border-radius: 4px;
            font-size: 18px;
            font-weight: 600;
            cursor: pointer;
            transition: background 0.2s;
        }
        button:hover {
            background: #2471a3;
        }
    </style>
</head>
<body>
    <div class="container">
        <h2>Conference Registration</h2>
        <form id="registrationForm" aria-labelledby="formTitle" novalidate>
```

```html
        <form id="registrationForm" aria-labelledby="formTitle" novalidate>
            <label for="name" aria-label="Full Name">Full Name</label>
            <input type="text" id="name" name="name" required aria-required="true" aria-describedby="nameError">
            <div id="nameError" class="error" aria-live="polite"></div>

            <label for="email" aria-label="Email Address">Email Address</label>
            <input type="email" id="email" name="email" required aria-required="true" aria-describedby="emailError">
            <div id="emailError" class="error" aria-live="polite"></div>

            <label for="phone" aria-label="Phone Number">Phone Number</label>
            <input type="tel" id="phone" name="phone" required aria-required="true" aria-describedby="phoneError" patter
            <div id="phoneError" class="error" aria-live="polite"></div>

            <label for="session" aria-label="Session Selection">Session Selection</label>
            <select id="session" name="session" required aria-required="true" aria-describedby="sessionError">
                <option value="">-- Select a session --</option>
                <option value="morning">Morning Session</option>
                <option value="afternoon">Afternoon Session</option>
                <option value="evening">Evening Session</option>
            </select>
            <div id="sessionError" class="error" aria-live="polite"></div>

            <button type="submit" aria-label="Submit Registration">Register</button>
        </form>
    </div>
    <script>
        const form = document.getElementById('registrationForm');
        const name = document.getElementById('name');
        const email = document.getElementById('email');
        const phone = document.getElementById('phone');
        const session = document.getElementById('session');
        const nameError = document.getElementById('nameError');
        const emailError = document.getElementById('emailError');
        const phoneError = document.getElementById('phoneError');
```

```
</div>
<script>
    const form = document.getElementById('registrationForm');
    const name = document.getElementById('name');
    const email = document.getElementById('email');
    const phone = document.getElementById('phone');
    const session = document.getElementById('session');
    const nameError = document.getElementById('nameError');
    const emailError = document.getElementById('emailError');
    const phoneError = document.getElementById('phoneError');
    const sessionError = document.getElementById('sessionError');

    function validateName() {
        if (!name.value.trim()) {
            nameError.textContent = 'Name is required.';
            return false;
        }
        nameError.textContent = '';
        return true;
    }

    function validateEmail() {
        const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
        if (!email.value.trim()) {
            emailError.textContent = 'Email is required.';
            return false;
        } else if (!emailPattern.test(email.value)) {
            emailError.textContent = 'Enter a valid email address.';
            return false;
        }
        emailError.textContent = '';
        return true;
    }
```

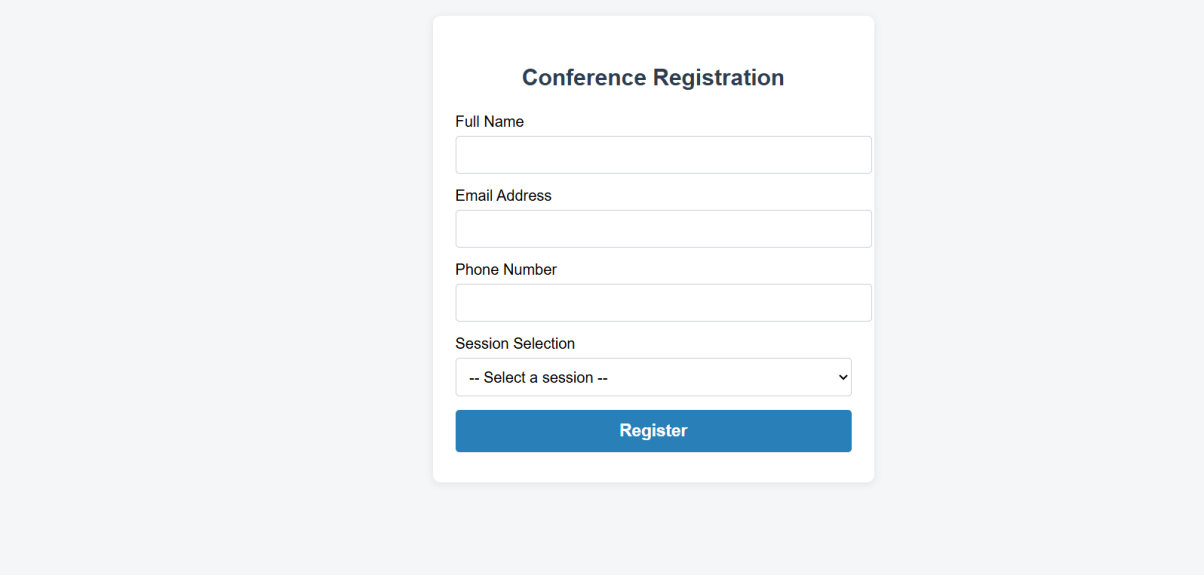
```
    function validatePhone() {
        const phonePattern = /^[0-9\-\+\s]{7,15}$/;
        if (!phone.value.trim()) {
            phoneError.textContent = 'Phone number is required.';
            return false;
        } else if (!phonePattern.test(phone.value)) {
            phoneError.textContent = 'Enter a valid phone number.';
            return false;
        }
        phoneError.textContent = '';
        return true;
    }

    function validateSession() {
        if (!session.value) {
            sessionError.textContent = 'Please select a session.';
            return false;
        }
        sessionError.textContent = '';
        return true;
    }

    name.addEventListener('input', validateName);
    email.addEventListener('input', validateEmail);
    phone.addEventListener('input', validatePhone);
    session.addEventListener('change', validateSession);

    form.addEventListener('submit', function(e) {
        let valid = true;
        if (!validateName()) valid = false;
        if (!validateEmail()) valid = false;
        if (!validatePhone()) valid = false;
        if (!validateSession()) valid = false;
        if (!valid) {
```

Output:


Conference Registration
Full Name
Email Address
Phone Number
Session Selection
-- Select a session --
Register

QUESTION 4)

CODE:

index4.html > ...

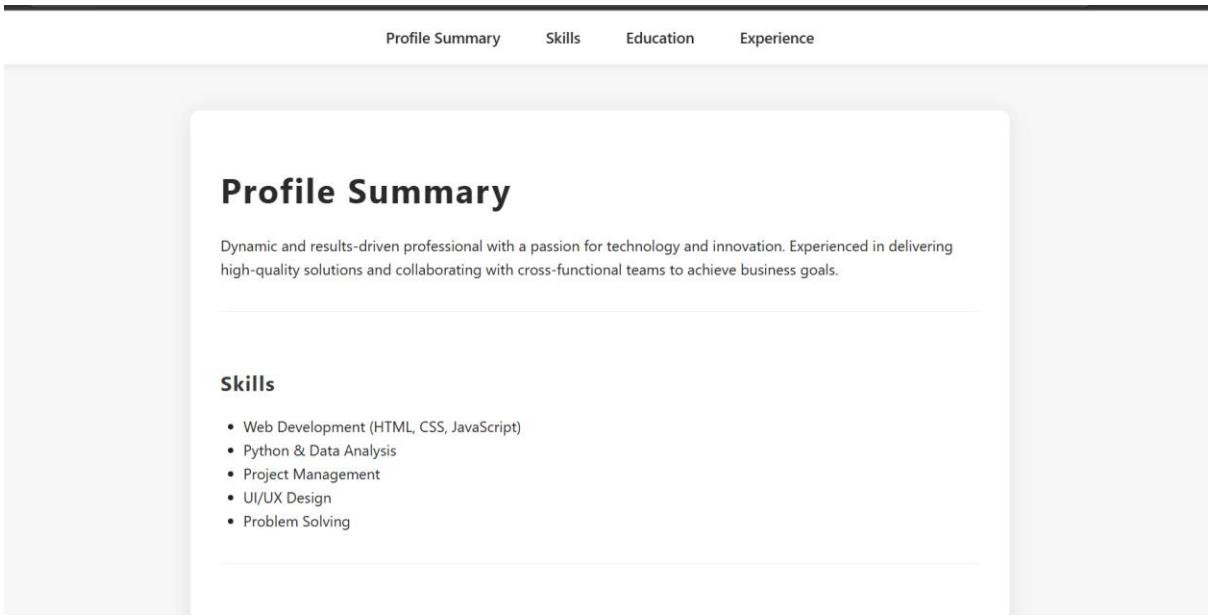
```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Resume Website</title>
  <style>
    /* Professional Typography */
    body {
      font-family: 'Segoe UI', 'Roboto', Arial, sans-serif;
      font-size: 18px;
      color: #222;
      background: #f7f7f7;
      margin: 0;
      line-height: 1.6;
    }
    h1, h2 {
      font-family: 'Montserrat', 'Segoe UI', Arial, sans-serif;
      font-weight: 700;
      letter-spacing: 0.05em;
      margin-bottom: 0.5em;
    }
    h1 {
      font-size: 2.5rem;
      color: #2a2a2a;
    }
    h2 {
      font-size: 1.5rem;
      color: #3a3a3a;
    }
    p, li {
      font-size: 1.1rem;
    }
    /* Smooth Scroll */
    html {
      scroll-behavior: smooth;
    }
```

```css
/* Navigation */
nav {
  background: #fff;
  box-shadow: 0 2px 8px rgba(0,0,0,0.07);
  padding: 1em 2em;
  position: sticky;
  top: 0;
  z-index: 10;
  display: flex;
  justify-content: center;
}
nav a {
  color: #2a2a2a;
  text-decoration: none;
  margin: 0 1.5em;
  font-weight: 600;
  transition: color 0.2s;
}
nav a:hover {
  color: #0077b6;
}
/* CSS Grid Layout */
.container {
  display: grid;
  grid-template-columns: 1fr;
  gap: 2.5em;
  max-width: 900px;
  margin: 3em auto;
  padding: 2em;
  background: #fff;
  border-radius: 12px;
  box-shadow: 0 4px 24px rgba(0,0,0,0.09);
}
```

```html
    <style>
      section {
        padding-bottom: 1em;
        border-bottom: 1px solid #ececec;
      }
      section:last-child {
        border-bottom: none;
      }
      ul {
        padding-left: 1.5em;
      }
      @media (max-width: 600px) {
        .container {
          padding: 1em;
        }
        nav {
          padding: 0.5em 1em;
        }
        h1 {
          font-size: 1.7rem;
        }
        h2 {
          font-size: 1.1rem;
        }
      }
    </style>
  </head>
  <body>
    <nav>
      <a href="#profile">Profile Summary</a>
      <a href="#skills">Skills</a>
      <a href="#education">Education</a>
      <a href="#experience">Experience</a>
    </nav>
    <div class="container">
```

```html
    <section id="profile">
      <h1>Profile Summary</h1>
      <p>Dynamic and results-driven professional with a passion for technology and innovation. Experienced in delivering
    </section>
    <section id="skills">
      <h2>Skills</h2>
      <ul>
        <li>Web Development (HTML, CSS, JavaScript)</li>
        <li>Python & Data Analysis</li>
        <li>Project Management</li>
        <li>UI/UX Design</li>
        <li>Problem Solving</li>
      </ul>
    </section>
    <section id="education">
      <h2>Education</h2>
      <ul>
        <li>B.Sc. in Computer Science, XYZ University (2018-2022)</li>
        <li>Certified Scrum Master (2023)</li>
      </ul>
    </section>
    <section id="experience">
      <h2>Experience</h2>
      <ul>
        <li>Software Developer, ABC Tech (2022-Present)</li>
        <li>Intern, DEF Solutions (2021-2022)</li>
      </ul>
    </section>
  </div>
</body>
</html>
```

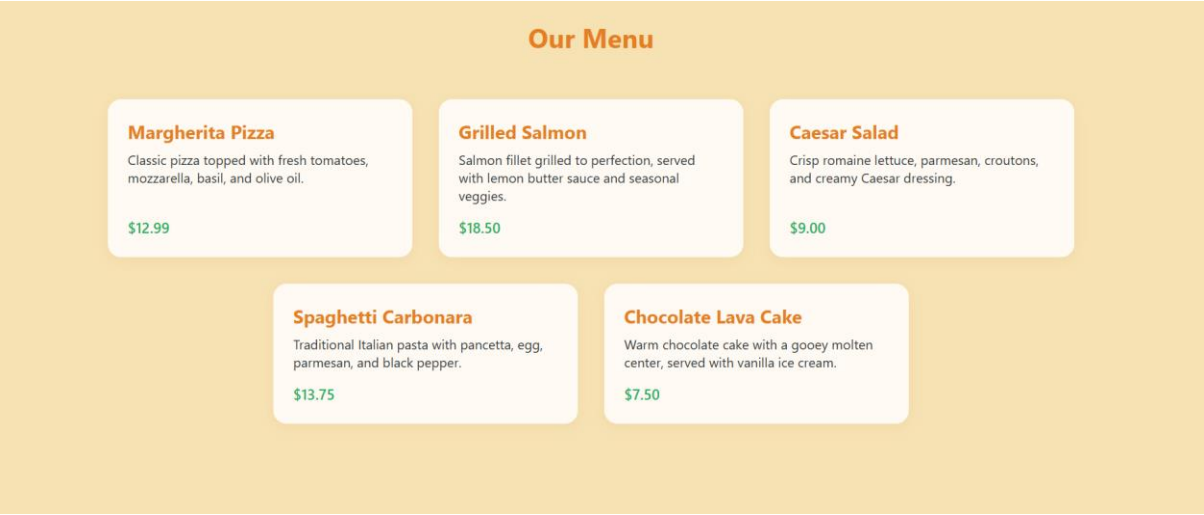
OUTPUT:


Profile Summary
Skills
Education
Experience
Profile Summary
Dynamic and results-driven professional with a passion for technology and innovation. Experienced in delivering high-quality solutions and collaborating with cross-functional teams to achieve business goals.
Skills
Web Development (HTML, CSS, JavaScript)
Python & Data Analysis
Project Management
UI/UX Design
Problem Solving

QUESTION 5)

CODE:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Menu</title>
    <style>
        /* AI-suggested food-friendly color palette */
        :root {
            --primary: #e67e22; /* carrot orange */
            --secondary: #f6e2b3; /* light beige */
            --accent: #27ae60; /* fresh green */
            --card-bg: #fffaf3; /* soft white */
            --text-main: #2d3436; /* dark gray */
            --hover-bg: #ffe5b4; /* peach */
        }
        body {
            background: var(--secondary);
            font-family: 'Segoe UI', Arial, sans-serif;
            color: var(--text-main);
            margin: 0;
            padding: 0;
        }
        .menu-container {
            display: flex;
            flex-wrap: wrap;
            gap: 2rem;
            justify-content: center;
            padding: 2rem;
        }
        .menu-card {
            background: var(--card-bg);
            border-radius: 16px;
            box-shadow: 0 4px 16px rgba(230, 103, 34, 0.08);
            width: 320px;
            max-width: 90vw;
            padding: 1.5rem;
```

```css
        .menu-card {
            transition: transform 0.2s, box-shadow 0.2s, background 0.2s;
            display: flex;
            flex-direction: column;
            align-items: flex-start;
        }
        .menu-card:hover {
            transform: translateY(-8px) scale(1.03);
            box-shadow: 0 8px 24px rgba(230, 103, 34, 0.18);
            background: var(--hover-bg);
        }
        .menu-title {
            font-size: 1.4rem;
            font-weight: bold;
            color: var(--primary);
            margin-bottom: 0.5rem;
        }
        .menu-desc {
            font-size: 1rem;
            margin-bottom: 1rem;
            color: var(--text-main);
        }
        .menu-price {
            font-size: 1.1rem;
            font-weight: 600;
            color: var(--accent);
            margin-top: auto;
        }
        @media (max-width: 700px) {
            .menu-container {
                gap: 1rem;
                padding: 1rem;
            }
            .menu-card {
```

```html
                padding: 1rem;
            }
        }
    </style>
</head>
<body>
    <h1 style="text-align:center;color:var(--primary);margin-top:2rem;">Our Menu</h1>
    <div class="menu-container">
        <div class="menu-card">
            <div class="menu-title">Margherita Pizza</div>
            <div class="menu-desc">Classic pizza topped with fresh tomatoes, mozzarella, basil, and olive oil.</div>
            <div class="menu-price">$12.99</div>
        </div>
        <div class="menu-card">
            <div class="menu-title">Grilled Salmon</div>
            <div class="menu-desc">Salmon fillet grilled to perfection, served with lemon butter sauce and seasonal veggies.
            <div class="menu-price">$18.50</div>
        </div>
        <div class="menu-card">
            <div class="menu-title">Caesar Salad</div>
            <div class="menu-desc">Crisp romaine lettuce, parmesan, croutons, and creamy Caesar dressing.</div>
            <div class="menu-price">$9.00</div>
        </div>
        <div class="menu-card">
            <div class="menu-title">Spaghetti Carbonara</div>
            <div class="menu-desc">Traditional Italian pasta with pancetta, egg, parmesan, and black pepper.</div>
            <div class="menu-price">$13.75</div>
        </div>
        <div class="menu-card">
            <div class="menu-title">Chocolate Lava Cake</div>
            <div class="menu-desc">Warm chocolate cake with a gooey molten center, served with vanilla ice cream.</div>
            <div class="menu-price">$7.50</div>
```